

Notes: Whited and Wu (RFS, 2006)

Financial Constraints Risk

Contents

1	Big picture	2
1.1	Incremental contribution of the paper	2
2	Data and measurement	2
2.1	Firm-level accounting sample	2
2.2	Structural financial constraints variables	3
2.3	KZ index comparison	3
2.4	Stock-return and portfolio data	4
3	Models and empirical specifications	4
3.1	Investment Euler-equation model	4
3.2	Shadow value and constraints index	4
3.3	Portfolio tests	5
3.4	Covariance tests	5
3.5	Cross-sectional Fama-MacBeth regressions	5
4	Identification strategy	5
4.1	What is identified?	5
4.2	Why the structural index is credible	5
4.3	How the asset-pricing tests separate constraints from size	6
4.4	Remaining limitations	6
5	Tables and figures	6
5.1	Table 1: Euler equation estimates	6
5.2	Tables 2 and 3: Summary statistics in the full sample and KZ sample	7
5.3	Table 4 and Figure 1: Portfolio characteristics and monthly cross-sorted financial constraints factor	8
5.4	Tables 5 and 6: Covariance tests of portfolios and preformation variances/covariances	9
5.5	Tables 7 and 8: Four-factor model and cross-sectional return regressions	10
6	Interpretive synthesis	11
7	Short summary	11
8	Conclusion	11
8.1	Core conclusion	11
8.2	Economic intuition	11
8.3	Final takeaway	12

1 Big picture

- **Question.** Do external finance constraints affect stock returns, and do financially constrained firms share a common source of systematic risk?
- **Core problem.** The common Kaplan-Zingales (KZ) index was built from a small selected sample and includes Tobin's q , which is prone to measurement error. The paper asks whether a structural investment model can produce a more credible financial-constraints index.
- **Main index.** Whited and Wu estimate a GMM investment Euler equation and use the fitted shadow value of scarce external finance as a financial constraints index.
- **Main validation.** Firms classified as constrained by the structural index are smaller, invest less despite investment opportunities, pay fewer dividends, receive less analyst coverage, and are much less likely to have bond ratings. By contrast, the KZ index classifies firms in ways that look inconsistent with financial constraints.
- **Asset-pricing result.** Returns of constrained firms move together. The resulting financial-constraints factor earns a positive but statistically weak average premium, and its variation is not fully explained by the Fama-French and momentum factors.
- **Cross-sectional result.** Fama-MacBeth regressions show that firms with higher financial-constraints-index values earn higher average returns. Once the constraints index is included, the usual negative size coefficient essentially disappears.
- **Economic takeaway.** Financial constraints are not only a corporate-investment friction; they appear to represent a source of common stock-return variation that helps explain part of the size effect.

1.1 Incremental contribution of the paper

- **Structural-index contribution.** The paper replaces the synthetic KZ index with an index derived from a dynamic investment model. This is a major improvement because the index is tied to the marginal shadow value of scarce external funds rather than to coefficients estimated in a small, nonexperimental sample.
- **Measurement contribution.** The paper avoids the standard investment- q measurement-error problem by estimating an Euler equation rather than regressing investment on noisy Tobin's q and cash flow.
- **Asset-pricing contribution.** Earlier work found common movement among constrained-firm returns but little priced risk. This paper shows that a structurally estimated constraints index generates common variation and positive cross-sectional return premia.
- **Size-effect contribution.** The paper shows that financial constraints can absorb the size effect in cross-sectional returns. This reframes the small-firm premium as potentially related to access to external finance rather than size per se.
- **Methodological contribution.** It connects structural corporate finance estimation with empirical asset pricing: first estimate firm-level financing frictions, then test whether those frictions are priced in time-series and cross-sectional returns.

2 Data and measurement

2.1 Firm-level accounting sample

The paper uses quarterly Standard & Poor's COMPUSTAT industrial files. The sample period for estimating the structural model is January 1975 to April 2001. The authors apply several filters:

- delete firm-years with missing data or nonpositive total assets, gross capital stock, or sales;

- delete observations with coding errors, such as short-term debt exceeding total debt;
- delete firms for which changes in capital stock cannot be reconciled with reported acquisitions, sales of capital goods, and depreciation;
- delete firms with mergers accounting for more than 15% of book assets;
- omit regulated utilities and financial firms, because the investment model is inappropriate for those sectors;
- require at least eight consecutive quarters of complete data;
- remove firms with more than two quarters of negative sales growth, because the goal is to study external finance constraints, not financial distress.

Interpretation: The sample design is meant to isolate operating firms whose investment decisions can be modeled with an Euler equation. The screens remove financial distress, regulatory structure, and severe accounting discontinuities.

2.2 Structural financial constraints variables

The shadow value of external finance is parameterized using firm characteristics that should predict financing frictions:

- CF : cash flow to total assets;
- $DIVPOS$: indicator for positive cash dividends;
- $TLTD$: long-term debt to total assets;
- $LNTA$: log of total assets;
- ISG : industry sales growth;
- SG : firm sales growth.

The preferred structural index, based on column 4 of Table 1, is

$$WW_{it} = -0.091CF_{it} - 0.062DIVPOS_{it} + 0.021TLTD_{it} - 0.044LNTA_{it} + 0.102ISG_{it} - 0.035SG_{it}.$$

Interpretation: Higher values of the index indicate a larger estimated shadow value of scarce external funds. The signs are economically intuitive: more cash flow, dividend payment, larger size, and stronger own sales growth lower constraints; leverage and high industry growth raise constraints.

2.3 KZ index comparison

The KZ index used by Lamont, Polk, and Saa-Requejo is

$$KZ_{it} = -1.001909CF_{it} + 3.139193TLTD_{it} - 39.36780TDIV_{it} - 1.314759CASH_{it} + 0.2826389Q_{it}.$$

Interpretation: The comparison is central. The KZ index is not just an alternative control variable; it is the incumbent measure of financial constraints. The paper argues that KZ poorly classifies firms in broad samples, while the structural index produces firm rankings consistent with standard economic intuition about constrained firms.

2.4 Stock-return and portfolio data

The asset-pricing tests merge the structural index with CRSP monthly stock returns from October 1975 to December 2001. Firms are independently sorted each quarter by size and by the structural constraints index into low, middle, and high categories. This creates nine portfolios:

$$SL, SM, SH, ML, MM, MH, BL, BM, BH,$$

where S, M, B denote small, medium, and big size groups, and L, M, H denote low, middle, and high financial constraints.

The factor-mimicking portfolio for financial constraints is

$$HIGHFC = (BH + MH + SH)/3, \quad LOWFC = (BL + ML + SL)/3, \quad FC = HIGHFC - LOWFC.$$

Interpretation: The double sort isolates financial constraints from size. Since constrained firms are often small, sorting only on constraints would mechanically confound the two. The FC factor compares constrained and unconstrained firms within size groups.

3 Models and empirical specifications

3.1 Investment Euler-equation model

The firm maximizes expected discounted dividends:

$$V_{i0} = E_{i0} \sum_{t=0}^{\infty} M_{0t} d_{it}.$$

Dividends equal profits net of adjustment costs, investment, and debt service, with capital evolving as

$$K_{i,t+1} = I_{it} + (1 - \delta_i)K_{it}.$$

The firm faces outside-finance constraints:

$$d_{it} \geq d_{it}^*, \quad B_{i,t+1} \leq B_{i,t+1}^*.$$

Interpretation: The Euler equation compares the marginal value of investing today versus postponing investment. If external finance is scarce, this intertemporal substitution is distorted by the shadow value of finance.

3.2 Shadow value and constraints index

Let the multiplier on the finance constraint be a function of firm characteristics. The GMM estimation produces parameter estimates for this multiplier, and the fitted value becomes the financial constraints index.

Interpretation: The index is structural in the sense that it comes from the model's first-order conditions rather than from reduced-form correlations between investment and cash flow. This is what lets the paper claim that the index measures the marginal cost of external finance rather than distress or size alone.

3.3 Portfolio tests

The paper forms nine size-by-constraint portfolios and linear combinations:

$$FC_t = HIGHFC_t - LOWFC_t.$$

The main time-series tests regress returns of portfolios or the FC factor on reference factors:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{p,M}MKT_t + \beta_{p,S}SMB_t + \beta_{p,H}HML_t + \beta_{p,U}UMD_t + \varepsilon_{p,t}.$$

Interpretation: A positive and unexplained intercept for FC means that known risk factors do not fully price the financial-constraints factor.

3.4 Covariance tests

The paper uses a Daniel-Titman style preformation covariance test. Firms switching into the constrained portfolio become more similar in constraint status as the formation date approaches. If constraints drive common return variation, variance of the switcher portfolio and covariance with the stayer portfolio should increase.

Interpretation: This test is designed to show that the FC factor is not merely an artifact of sorting on industries, regions, or other correlated characteristics. It asks whether the common variation intensifies exactly as constraint status becomes more similar.

3.5 Cross-sectional Fama-MacBeth regressions

The paper estimates monthly cross-sectional regressions of stock returns on firm characteristics and reports time-series means of coefficients:

$$R_{i,t} - R_{f,t} = a_t + b_{1,t}Size_{i,t-1} + b_{2,t}BM_{i,t-1} + b_{3,t}Momentum_{i,t-1} + b_{4,t}WW_{i,t-1} + u_{i,t}.$$

Interpretation: The coefficient on WW measures whether more constrained firms earn higher returns conditional on size, book-to-market, and momentum. If the size coefficient disappears after adding WW , then financial constraints explain part of the size premium.

4 Identification strategy

4.1 What is identified?

The paper does not estimate a randomized treatment effect. It identifies a structurally motivated firm characteristic, the shadow value of external finance, and then asks whether that characteristic predicts common variation and expected returns.

Interpretation: The identification problem is measurement: how to measure financial constraints without relying on endogenous cash-flow sensitivity, noisy q , or coefficients imported from a small sample.

4.2 Why the structural index is credible

- The Euler equation is estimated on a broad panel rather than on a small set of hand-classified firms.

- GMM overidentifying restrictions are not rejected for the preferred specifications.
- The fitted index ranks firms in ways consistent with external finance constraints: low dividends, low analyst coverage, little bond rating access, small size, and low investment despite growth opportunities.
- The index is not just another expression of KZ: the cross-sectionally demeaned correlation with KZ is close to zero in the full sample.

Interpretation: This combination of structural estimation and external validation is the paper’s main identification argument for the index.

4.3 How the asset-pricing tests separate constraints from size

- Portfolios are sorted independently by size and the constraints index.
- The FC factor is long constrained firms and short unconstrained firms within size classes.
- Factor regressions include market, SMB, HML, and momentum factors.
- Cross-sectional regressions include size directly and show that the size coefficient vanishes when the constraints index enters.

Interpretation: Since constrained firms are often small, these design choices are necessary. Without them, the constraints premium could be a mislabeled size premium.

4.4 Remaining limitations

- The index depends on the correctness of the investment Euler-equation model and the parameterization of the finance multiplier.
- The sample excludes private firms and venture-backed firms, even though they may be severely finance constrained.
- The time-series FC factor premium is positive but not statistically significant.
- The analysis shows that constraints are priced characteristics and common factors, but it does not prove which macro shocks drive the factor.

Interpretation: The paper is strongest on measurement and cross-sectional pricing. It is more cautious on whether the FC factor itself earns a precisely estimated standalone time-series premium.

5 Tables and figures

5.1 Table 1: Euler equation estimates

Read: Table 1 reports GMM estimates of the investment Euler equation. The preferred column yields the structural index

$$-0.091CF - 0.062DIVPOS + 0.021TLTD - 0.044LNTA + 0.102ISG - 0.035SG.$$

Cash flow, dividends, size, and own sales growth reduce the index; leverage and industry sales growth increase it.

Interpretation: The signs make economic sense. Firms with high internal cash flow, dividend payments, and large scale behave as less constrained. Firms with high leverage and high industry growth opportunities behave as more constrained because they need external finance when external funds are costly.

Economic message: Table 1 is the foundation for the paper: it converts a dynamic investment model into a firm-quarter constraints index that can be taken to asset-pricing tests.

Table 1
Euler equation estimates

	1	2	3	4	5
α_1	0.534 (0.190)	0.655 (0.174)	0.608 (0.179)	0.652 (0.167)	0.701 (0.152)
α_2	-0.354 (0.349)	-0.402 (0.399)	-0.490 (0.261)	-0.442 (0.228)	-0.437 (0.256)
μ	0.967 (0.012)	1.011 (0.019)	1.018 (0.024)	1.019 (0.023)	1.018 (0.023)
<i>CF</i>	-0.079 (0.034)	-0.063 (0.026)	-0.072 (0.025)	-0.091 (0.031)	-0.098 (0.031)
<i>DIVPOS</i>	-0.054 (0.022)	-0.062 (0.034)	-0.046 (0.021)	-0.062 (0.029)	-0.073 (0.030)
<i>TLTD</i>	0.026 (0.013)	0.011 (0.008)	0.025 (0.008)	0.021 (0.011)	0.013 (0.007)
<i>LNTA</i>	-0.077 (0.024)	-0.120 (0.030)	-0.040 (0.028)	-0.044 (0.023)	-0.054 (0.023)
<i>ISG</i>	0.121 (0.104)	0.117 (0.105)	0.066 (0.057)	0.102 (0.052)	0.085 (0.057)
<i>SG</i>	-0.031 (0.011)	-0.050 (0.025)	-0.024 (0.011)	-0.035 (0.023)	
<i>NA</i>	-0.004 (0.002)	-0.007 (0.004)	-0.019 (0.090)		
<i>CASH</i>	-0.001 (0.002)	-0.009 (0.011)			
<i>IDAR</i>	-0.011 (0.042)				
<i>MKT</i>	-0.539 (0.232)	-0.227 (0.685)	-0.780 (0.646)	-0.556 (0.318)	-0.276 (0.144)
<i>SMB</i>	1.285 (0.665)	1.033 (0.880)	1.020 (0.324)	1.083 (0.735)	0.936 (0.493)
<i>HML</i>	1.121 (0.492)	1.069 (1.500)	0.412 (0.652)	0.944 (0.770)	0.906 (0.556)
<i>J-Test</i>	0.216	0.242	0.229	0.193	0.024
<i>L-Test</i>		0.397	0.499	0.595	0.062

Calculations are based on a sample of nonfinancial firms from the quarterly 2002 COMPUSTAT industrial files. The sample period is January, 1975 to April, 2001. The model is given by equation (9). Nonlinear GMM estimation is done on the model in first differences with twice lagged instruments. α_1 and α_2 are adjustment cost parameters, and μ is a mark-up. *CF* is the ratio of cash flow to total assets; *DIVPOS* is an indicator that takes the value of one if the firm pays cash dividends; *TLTD* is the ratio of the long-term debt to total assets; *LNTA* is the natural log of total assets; *ISG* is the firm's 3-digit industry sales growth; *SG* is firm sales growth; *NA* is the number of analysts following the firm, as reported by I/B/E/S; *CASH* is the ratio of liquid assets to total assets; and *IDAR* is the firm's 3-digit industry debt-to-assets ratio. *MKT*, *SMB*, and *HML* are the Fama-French factors on market, size and book-to-market. Standard errors are reported in parentheses. The *p*-values of the *J*-test and *L*-test on model specification are reported in the last two rows.

Table 1: Euler equation estimates.

5.2 Tables 2 and 3: Summary statistics in the full sample and KZ sample

Read: Table 2 compares firm characteristics across quartiles of the structural index and the KZ index in the full sample. Table 3 repeats the exercise in the original KZ-related sample.

- Structural-index constrained firms invest less despite higher Tobin's q and industry growth.
- They pay fewer dividends, have less analyst coverage, and almost no bond ratings.
- KZ-constrained firms in the broad sample look anomalous: they often have more analyst coverage, more bond ratings, and investment rising rapidly with q .
- In the original KZ sample, KZ works better, illustrating the danger of transporting coefficients from one nonexperimental sample to another.

Interpretation: These tables validate the structural index and explain why the paper departs from KZ. The structural index behaves like an external-finance-constraints measure in the broad sample; KZ does not.

Economic message: Measurement of constraints is not a technical detail. The asset-pricing conclusion depends on correctly identifying which firms are actually constrained.

Table 2
Summary statistics: full sample

	Least constrained		Most constrained	
Firms sorted by structural index				
Investment/assets	0.095	0.095	0.088	0.078
Cash flow/assets	0.110	0.102	0.092	0.053
Total assets	1151.436	194.547	66.352	22.367
Debt assets	0.221	0.189	0.167	0.145
Positive dividends	0.549	0.292	0.119	0.031
Industry sales growth	-0.018	0.013	0.035	0.158
Sales growth	0.043	0.054	0.031	0.028
Cash/assets	0.081	0.105	0.122	0.128
Industry debt/assets	0.212	0.193	0.179	0.173
Number of analysts	1.414	0.871	0.459	0.123
Bond rating	0.231	0.074	0.017	0.003
Tobin's q	2.017	2.102	2.077	2.197
Structural index	0.588	0.666	0.724	0.803
Kaplan-Zingales index	1.094	1.004	0.935	0.944
Firms sorted by Kaplan-Zingales index				
Investment/assets	0.063	0.083	0.096	0.114
Cash flow/assets	0.144	0.095	0.074	0.045
Total assets	166.889	354.783	463.596	449.415
Debt assets	0.030	0.110	0.210	0.373
Positive dividends	0.279	0.259	0.249	0.205
Industry sales growth	0.039	0.050	0.052	0.047
Sales growth	0.029	0.040	0.048	0.059
Cash/assets	0.216	0.095	0.066	0.058
Industry debt/assets	0.147	0.175	0.202	0.234
Number of analysts	0.496	0.737	0.832	0.802
Bond rating	0.012	0.034	0.082	0.196
Tobin's q	1.926	2.033	2.041	2.392
Structural index	0.712	0.694	0.684	0.689
Kaplan-Zingales index	0.254	0.765	1.148	1.808

Calculations are based on a sample of nonfinancial firms from the quarterly 2002 COMPUSTAT industrial files. The sample period is January, 1975 to April, 2001. Investment/assets, sales/assets, and cash flow/assets are expressed at an annual rate. Industry sales growth is defined at the three-digit SIC level. Total assets are expressed in millions of 1997 dollars. The denominator of Tobin's q is the book value of total assets. The numerator is the book value of total assets minus the book value of equity minus balance sheet deferred taxes plus the market value of equity.

(a) Table 2: Summary statistics, full sample.

Table 3
Summary statistics: KZ sample

	Least constrained		Most constrained	
Firms sorted by structural index				
Investment/assets	0.098	0.096	0.076	0.092
Cash flow/assets	0.106	0.104	0.108	0.041
Total assets	3009.205	487.919	227.920	98.932
Debt assets	0.116	0.121	0.143	0.180
Positive dividends	0.342	0.335	0.082	0.008
Industry sales growth	-0.015	0.024	0.018	0.139
Sales growth	0.035	0.047	0.017	0.087
Cash/assets	0.131	0.097	0.082	0.067
Industry debt/assets	0.124	0.144	0.152	0.168
Number of analysts	2.723	1.959	1.706	0.997
Bond rating	0.444	0.070	0.004	0.012
Tobin's q	2.786	2.437	2.458	2.086
Structural index	0.552	0.619	0.665	0.723
Kaplan-Zingales index	0.938	0.896	1.011	1.120
Firms sorted by Kaplan-Zingales index				
Investment/assets	0.063	0.098	0.099	0.101
Cash flow/assets	0.136	0.100	0.084	0.041
Total assets	1650.666	787.019	683.659	694.077
Debt assets	0.036	0.081	0.169	0.270
Positive dividends	0.335	0.202	0.097	0.132
Industry sales growth	0.040	0.054	0.046	0.066
Sales growth	0.042	0.034	0.037	0.073
Cash/assets	0.119	0.114	0.075	0.068
Industry debt/assets	0.108	0.137	0.169	0.169
Number of analysts	1.668	2.052	1.788	1.864
Bond rating	0.125	0.109	0.121	0.171
Tobin's q	1.941	2.499	2.379	2.885
Structural index	0.620	0.637	0.645	0.660
Kaplan-Zingales index	0.424	0.794	1.110	1.606

Calculations are based on a sample of nonfinancial firms from the quarterly 2002 COMPUSTAT industrial files. The sample period is January, 1975 to April, 2001. Investment/assets, sales/assets, and cash flow/assets are expressed at an annual rate. Industry sales growth is defined at the three-digit SIC level. Total assets are expressed in millions of 1997 dollars. The denominator of Tobin's q is the book value of total assets. The numerator is the book value of total assets minus the book value of equity minus balance-sheet deferred taxes plus the market value of equity.

(b) Table 3: Summary statistics, KZ sample.

5.3 Table 4 and Figure 1: Portfolio characteristics and monthly cross-sorted financial constraints factor

Read: Table 4 reports returns and characteristics for nine size-by-constraints portfolios. Figure 1 plots cumulative returns of the value-weighted and equal-weighted financial constraints factors.

- Constrained firms are disproportionately small.
- The FC portfolio earns positive average returns: 0.18 for value-weighted and 0.23 for equal-weighted monthly excess returns, but with modest t -statistics.
- Figure 1 shows that the FC factor is countercyclical: its cumulative returns often rise around recessions and fall during expansions.

Interpretation: Table 4 documents that constrained portfolios have higher returns but also strong overlap with small size. Figure 1 shows that the factor behaves like a macro-finance variable, not just a static firm characteristic.

Economic message: Financial constraints are particularly costly in bad aggregate states. This cyclicalitly is why constraints can be a source of priced risk.

Table 4
Portfolio characteristics and returns

	Category label	Number of firms	Value weighted				Equal weighted			
			Excess returns	D/A	B/M	Size	Excess returns	D/A	B/M	Size
Small-cap firms										
Low index	SL	37	0.89	0.54	1.92	0.03	0.87	0.56	2.18	0.03
Middle index	SM	85	0.66	0.38	1.30	0.03	0.82	0.43	1.52	0.03
High index	SH	349	0.83	0.23	0.89	0.03	1.15	0.28	1.12	0.02
Mid-cap firms										
Low index	ML	70	0.65	0.38	1.27	0.10	0.65	0.40	1.31	0.09
Middle index	MM	77	0.81	0.22	0.82	0.10	0.78	0.23	0.85	0.09
High index	MH	89	0.74	0.13	0.57	0.09	0.75	0.14	0.58	0.08
Large-cap firms										
Low index	BL	367	0.71	0.19	0.74	19.09	0.69	0.24	0.80	2.84
Middle index	BM	74	0.96	0.11	0.46	0.76	0.97	0.13	0.53	0.37
High index	BH	33	1.23	0.10	0.41	1.03	1.02	0.09	0.42	0.36
HIGHFC			0.93	0.15	0.63	0.38	0.97	0.17	0.71	0.15
LOWFC			0.75	0.37	1.31	6.41	0.74	0.40	1.43	0.98
FC			0.18	-0.22	-0.68	-6.03	0.23	-0.23	-0.72	-0.83
t-stat of FC			0.95				1.32			

B/M, book-to-market ratio; D/A, debt-to-assets ratio; FC, financial constraints factor; BH large size/high index; BL large size/low index; BM, large size/middle index; MH, medium size/high index; ML, medium size/low index; MM, medium size/middle index; SH, small size/high index; SL, small size/low index; SM, small size/middle index.

This table reports summary statistics for nine value-weighted and nine equal-weighted portfolios formed by rankings of the market capitalization and the structural financial constraints index. The rankings are performed independently such that each portfolio contains firms that are both in a given size category and a given financial constraints category. Small-cap firms are firms that are in the bottom 40% of the sample in a given quarter sorted on market capitalization. Mid-cap firms are firms that are in the middle 20% of the sample. Large-cap firms are firms that are in the top 40% of the sample. Similarly, low, middle, and high index are firms that are in the bottom 40%, the middle 20%, and the top 40% of the sample sorted by the structural financial constraints index in a given quarter. $HIGHFC = (BH + MH + SH)/3$, $LOWFC = (BL + ML + SL)/3$, $FC = HIGHFC - LOWFC$. We report the sample mean of each portfolio's monthly returns in excess of one-month Treasury Bill yields in percentage. We also calculate average number of firms in each portfolio, D/A, B/M, and market capitalization in billions of dollars (size) by averaging over the entire sample period. The sample period is from October, 1975 to December, 2001.

(a) Table 4: Portfolio characteristics and returns.



Figure 1
Monthly cross-sorted financial constraints factor

This figure plots the value-weighted and equal-weighted cross-sorted financial constraints factors. Based on independent sorts of the top 40, middle 20, and bottom 40% of size, and the financial constraints index, we classify all firms into one of nine groups: small size/low index (SL), small size/middle index (SM), small size/high index (SH), medium size/low index (ML), medium size/middle index (MM), medium size/high index (MH), large size/low index (BL), large size/middle index (BM), and large size/high index (BH). We form portfolios based on these sorts, using the quarterly financial constraints index estimated in our generalized method of moments (GMM) framework. Subsequent equal-weighted and value-weighted average monthly returns on these portfolios are calculated with Center for Research in Security Prices (CRSP) monthly data. Average constrained portfolio returns are computed as the mean of the SH, MH, and BH portfolio returns. Average unconstrained portfolio returns are computed as the mean of the SL, ML, and BL portfolio returns. The financial constraints factor is the average constrained portfolio returns minus the average unconstrained portfolio returns. We indicate the beginning and end of NBER recessions with vertical dashed and solid lines. The sample period is from October, 1975 to December, 2001.

(b) Figure 1: Monthly cross-sorted financial constraints factor.

5.4 Tables 5 and 6: Covariance tests of portfolios and preformation variances/covariances

Read: Table 5 shows that constrained portfolio returns load significantly on the FC factor after controlling for market and size reference portfolios. Table 6 implements a preformation test: as switchers become more similar in constraint status, variance and covariance increase.

- In Table 5, high-index portfolios generally load positively on the FC factor.
- In Table 6, $FC(\text{switch})$ variance rises from 15.48 at $t - 5$ to 25.61 at t , and covariance with $FC(\text{stay})$ rises from 3.79 to 12.39.
- The coefficient on $FC(\text{stay})$ rises from 0.28 to 0.57.

Interpretation: These tables are the paper's strongest evidence of common return variation associated with financial constraints. The Daniel-Titman style test is important because it is designed to rule out spurious commonality from other portfolio characteristics.

Economic message: Constrained firms do not merely have high average returns; their returns move together in a way that intensifies when constraint status becomes more similar.

Table 5
Covariance tests of portfolio

	Regression results					Variable definitions		
	Constant	BIG	SMALL	FC	R ²	BIG	SMALL	FC
Small-cap firms								
Low index (SL)	0.00 (0.66)	0.42 (5.25)	0.71 (10.11)	-0.10 (-1.60)	0.68	(BM + BL + MM + ML) ₃₄	SM	(BH + MH - BL - ML) ₃₂
Mid-index (SM)	-0.00 (-0.72)	0.56 (11.25)	0.34 (10.10)	0.13 (2.92)	0.76	(BM + BL + MM + ML) ₃₄	SL	(BH + MH - BL - ML) ₃₂
High index (SH)	0.00 (0.01)	0.43 (11.02)	0.50 (15.93)	0.29 (9.56)	0.88	(BM + BL + MM + ML) ₃₄	SM + SL ₃₂	(BH + MH - BL - ML) ₃₂
Mid-cap firms								
Low index (ML)	-0.00 (-1.76)	0.64 (16.00)	0.43 (14.09)	0.01 (0.591)	0.88	(BM + BL + MM) ₃₃	(SM + SL) ₃₂	(BH + SH - BL - SL) ₃₂
Mid-index (MM)	-0.00 (-0.23)	0.74 (21.76)	0.27 (10.06)	0.18 (6.83)	0.90	(BM + BL + ML) ₃₃	SM + SL ₃₂	(BH + SH - BL - SL) ₃₂
High index (MH)	-0.00 (-1.55)	0.75 (18.84)	0.31 (9.84)	0.43 (14.30)	0.91	(BM + BL + MM + ML) ₃₄	SM + SL ₃₂	(BH + SH - BL - SL) ₃₂
Large-cap firms								
Low index (BL)	0.00 (1.25)	0.83 (15.44)	-0.20 (-4.17)	-0.03 (-0.70)	0.66	(BM + MM + ML) ₃₃	(SM + SL) ₃₂	(MH + SH - ML - SL) ₃₂
Mid-index (BM)	0.00 (1.49)	1.04 (22.32)	0.00 (0.11)	0.29 (7.62)	0.85	(BL + MM + ML) ₃₃	SM + SL ₃₂	(MH + SH - ML - SL) ₃₂
High index (BH)	0.00 (1.49)	0.86 (9.99)	0.24 (1.34)	0.76 (9.31)	0.68	(BM + BL + MM + ML) ₃₄	SM + SL ₃₂	(MH + SH - ML - SL) ₃₂

This table reports regression results of nine value-weighted portfolios described in Table 4. We regress the excess return on each portfolio on three reference portfolios: a market proxy, a size factor proxy, and the financial constraints factor (FC). We construct our size and market proxies using the portfolios in Table 4 as follows. The proxy for overall market is the return on a portfolio of less-constrained medium-size and large firms, BIG = (ML + MM + BL + BM)₃₄, in excess of one-month Treasury Bill yield. The proxy for size is the return on a portfolio of less-constrained small firms, SMALL = (SL + SM)₃₂, in excess of one-month Treasury Bill yield. FC is defined in Table 4. In each regression, we omit the portfolio that is the dependent variable from the construction of the portfolios that constitute the regression's independent variables. In the case of FC, we also omit the matching portfolio on the short side. *T*-statistics are in parentheses. The sample period is from October, 1975 to December, 2001.

(a) Table 5: Covariance tests of portfolios.

Table 6
Preformation quarterly return variances and covariances

	FC(switch)			FC(stay)		FC(switch) and FC(stay) regression results		
	Percent switching	Variance	Standard deviation	Variance	Standard deviation	Covariance	Coefficient on FC(stay)	R ²
$t - 5$	100	15.48	3.93	36.01	6.00	3.79	0.28 (5.68)	0.09
$t - 4$	95	16.28	4.03	39.17	6.25	7.69	0.41 (7.62)	0.20
$t - 3$	91	20.03	4.47	38.30	6.18	9.49	0.50 (9.24)	0.21
$t - 2$	65	20.41	4.51	39.07	6.25	7.23	0.43 (7.68)	0.15
$t - 1$	37	21.39	4.68	30.21	5.49	7.80	0.46 (8.07)	0.17
t	0	25.63	5.06	27.68	5.26	12.39	0.57 (9.61)	0.22

This table presents the time-series properties of the return on two portfolios, FC(switch) and FC(stay). It is constructed in a similar fashion to Table 4 of Lamont et al. (2001), except that we use the structural financial constraint index instead of the KZ index and we use quarterly accounting data instead of the annual data. The portfolios are constructed from the sample of all firms that are in the financial constraint portfolio in quarter t (so that they are in the top third or bottom third of all firms ranked by the structural financial constraint index at the end of quarter $t - 1$) and which also have data available to construct the structural financial constraint index in quarter $t - 6$. FC(stay) goes long on firms that are

(b) Table 6: Preformation return variances and covariances.

5.5 Tables 7 and 8: Four-factor model and cross-sectional return regressions

Read: Table 7 regresses the FC factor on market, SMB, HML, and momentum factors. Table 8 runs Fama-MacBeth cross-sectional return regressions on size, book-to-market, momentum, and the FC index.

- Table 7 shows that the four-factor model explains only 37–50% of full-sample FC factor variation, and intercepts remain economically and statistically meaningful in several specifications.
- The FC factor loads positively on SMB and momentum and negatively on HML; it is not reducible to those factors.
- Table 8 shows that the financial constraints index has a positive and significant coefficient in Models 6 and 7.
- Adding the constraints index changes the size coefficient from negative to essentially zero.

Interpretation: Table 7 shows that FC is not simply a repackaged Fama-French-momentum factor. Table 8 shows that the firm-level constraints characteristic predicts returns and absorbs the size effect.

Economic message: The paper's central asset-pricing message is here: financial constraints help explain expected returns and may be an economic source behind the empirical size premium.

Table 7
Relating the financial constraints factor to the four-factor model

Dependent variable	Constant	Market	SMB	HML	Momentum	R ²
Full Sample						
Value-weighted FC factor	0.0052 (3.37)	-0.1977 (-5.44)	0.3226 (6.34)	-0.5648 (-11.74)	0.2608 (6.77)	42.91%
Equal-weighted FC factor	0.0020 (1.98)	-0.1724 (-5.08)	0.3478 (7.23)	-0.4001 (-7.89)	0.2608 (6.77)	50.42%
Equal-weighted FC factor	0.0004 (4.33)	-0.2008 (-6.50)	0.1453 (2.93)	-0.5617 (-11.95)	0.2608 (6.77)	37.62%
Equal-weighted FC factor	0.0027 (1.92)	-0.2019 (-6.23)	0.1740 (3.55)	-0.3752 (-7.73)	0.2608 (6.77)	48.34%
October, 1975–November, 1988						
Value-weighted FC factor	0.0007 (0.34)	-0.0012 (-1.08)	0.2963 (3.73)	-0.1889 (-2.35)	0.1809 (3.05)	12.20%
Equal-weighted FC factor	-0.0004 (-0.21)	-0.0747 (-1.59)	0.2605 (3.34)	-0.1507 (-1.90)	0.1809 (3.05)	17.22%
Equal-weighted FC factor	0.0015 (0.71)	-0.0777 (-1.62)	0.1587 (1.96)	-0.2151 (-2.65)	0.1809 (3.05)	6.79%
Equal-weighted FC factor	-0.0052 (-0.09)	-0.1225 (-2.40)	0.1033 (1.36)	-0.1585 (-2.05)	0.2608 (4.64)	18.23%
December, 1988–December, 2001						
Value-weighted FC factor	0.0073 (3.40)	-0.2197 (-5.97)	0.2619 (4.07)	-0.7490 (-12.57)	0.2665 (4.70)	62.19%
Equal-weighted FC factor	0.0030 (1.36)	-0.2142 (-3.90)	0.3628 (5.67)	-0.4981 (-6.45)	0.2665 (4.70)	67.00%
Equal-weighted FC factor	0.0093 (4.63)	-0.2748 (-7.42)	0.0675 (1.11)	-0.7425 (-13.23)	0.2665 (4.70)	59.16%
Equal-weighted FC factor	0.0052 (2.51)	-0.2715 (-5.27)	0.1659 (2.78)	-0.4972 (-6.86)	0.2604 (4.96)	64.73%

This table reports the estimates of models linking the value-weighted and equal-weighted financial constraints factor to the three Fama–French factors and the momentum factor. *t*-statistics are reported below the coefficients in parentheses. The sample period is from October, 1975 to December, 2001.

(a) Table 7: Financial constraints factor and the four-factor model.

Table 8
Cross-sectional regression of returns on firm characteristics

Specification	Size	B/M	Size/BM	Momentum	FC index
Full sample					
Model 1	-0.0052 (-1.52)				
Model 2		0.0046 (5.10)			
Model 3	-0.0043 (-1.31)	0.0045 (5.08)			
Model 4	-0.0042 (-1.35)	0.0050 (6.04)		0.0278 (1.95)	
Model 5	-0.0056 (-1.54)	0.0050 (6.14)	0.0001 (0.55)	0.0273 (1.92)	
Model 6	0.0002 (0.58)	0.0051 (6.12)		0.0277 (1.99)	0.0349 (3.01)
Model 7	0.0002 (0.59)	0.0050 (6.10)	0.0001 (0.46)	0.0272 (1.97)	0.0363 (3.17)
October, 1975–November, 1988					
Model 6	0.0000 (0.07)	0.0067 (5.50)		0.0420 (2.35)	0.0204 (1.39)
Model 7	0.0000 (0.10)	0.0066 (5.53)	0.0000 (0.39)	0.0427 (2.42)	0.0191 (1.31)
December, 1988–December, 2001					
Model 6	0.0000 (2.30)	0.0036 (3.11)		0.0133 (0.62)	0.0498 (2.77)
Model 7	0.0000 (2.22)	0.0034 (3.06)	0.0000 (1.77)	0.0118 (0.54)	0.0537 (3.03)

B/M, book-to-market ratio; FC, financial constraints. This table reports the results for the month-by-month cross-sectional regressions of firm excess returns on firm characteristics such as size, book-to-market ratio, momentum and financial constraints. Excess returns are computed in excess of one-month Treasury Bill yields. Size is measured by market capitalization in billions of dollars. Momentum is measured by prior six-month mean return excluding the latest month. Financial constraints are measured by the structural financial constraints index. We also define an interaction between size and book-to-market: size/BM equals market capitalization divided by the book-to-market ratio. We use the Fama–Macbeth technique to compute the means of the time series of regression coefficients. The time-series *t*-statistics are reported in parentheses. The sample period is

(b) Table 8: Cross-sectional regression of returns on firm characteristics.

6 Interpretive synthesis

- **Measurement comes first.** The paper’s asset-pricing claims depend on building a credible constraints measure. The structural index passes validation tests that KZ fails in the broad sample.
- **Constraints as a factor.** The FC factor has common variation not spanned by market, size, value, and momentum factors.
- **Constraints as a characteristic.** In cross-sectional regressions, constrained firms earn higher returns. This result is stronger than the time-series average premium of the FC factor itself.
- **Relationship to size.** The size effect weakens sharply once the constraints index is included, suggesting that part of the size premium may reflect exposure to financing frictions.
- **Limits.** The paper identifies a priced characteristic and common factor, but it does not fully identify the macro shocks that produce the FC factor or extend to private firms.

7 Short summary

- Whited and Wu construct a structural index of external finance constraints from a GMM investment Euler equation.
- The index is designed to measure the shadow value of scarce external funds.
- Constrained firms under the WW index are small, low-dividend, low-analyst-coverage, low-bond-rating, and underinvesting relative to growth opportunities.
- The KZ index performs poorly in the broad sample and has low correlation with the WW index.
- The paper constructs an FC factor by sorting firms on size and the constraints index.
- Constrained firms’ returns co-move, and the FC factor is countercyclical.
- The FC factor has positive but imprecise average returns, and much of its variation is not explained by standard factors.
- Cross-sectional regressions show higher returns for more constrained firms.
- Adding the constraints index removes the size effect.
- The paper’s incremental contribution is a structural financial constraints index that connects corporate finance frictions to asset-pricing premia.

8 Conclusion

8.1 Core conclusion

Whited and Wu show that firm-level external finance constraints are reflected in stock returns. Their structural index identifies firms that look plausibly constrained and shows that these firms share common return variation and earn higher cross-sectional returns.

8.2 Economic intuition

External finance constraints make firms more exposed to aggregate shocks because adverse states reduce collateral values, internal funds, and access to external capital. If constrained firms are forced to cut investment more sharply in bad times, their equity returns should covary with a state variable that investors care about. The FC factor is an empirical proxy for that state variable.

8.3 Final takeaway

The paper's enduring lesson is that measuring financial constraints carefully matters. A poorly transported index can generate misleading classifications, while a structural index can reveal that financing frictions are related to systematic risk and expected returns.